

The canonical bundle map of an essential $C([0, 1])$ -module

A counterexample to the injectivity question in arXiv:1005.4561

Autonomous solve-first investigation, lane 12

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Abstract

Leung, Ng, and Wong ask whether the canonical map from every essential Banach $C(\Omega)$ -module to its associated section module is injective when Ω is a compact subset of $\mathbb{R}$. The answer is no, already for $\Omega = [0, 1]$. With its usual multiplication action, $F = L^1([0, 1])$ is an essential Banach $C([0, 1])$ -module, but every canonical fiber F/K_ω^F is zero. Consequently the canonical map is identically zero. The same phenomenon occurs for every L^p -module over an atomless finite regular measure, $1 \leq p < \infty$.

1 The question

Let F be an essential Banach $C(\Omega)$ -module. For $\omega \in \Omega$, the source defines

$$K_\omega = \{a \in C(\Omega) : a(\omega) = 0\}, \quad K_\omega^F = \overline{FK_\omega},$$

and the canonical fiber $\Xi_\omega^F = F/K_\omega^F$. The canonical map is

$$\iota_F : F \longrightarrow \Gamma_0(\Xi^F), \quad \iota_F(x)(\omega) = x + K_\omega^F. \quad (1)$$

The closure of its image is denoted by $\tilde{F}$.

Remark 4.10(b) of the source asks whether (1) is injective for every compact $\Omega \subseteq \mathbb{R}$ and every essential Banach $C(\Omega)$ -module F [1, p. 13]. Figure 1 reproduces the exact passage.

2 A maximally noninjective example

Theorem 1. *Let $\Omega = [0, 1]$, let $F = L^1([0, 1])$ with respect to Lebesgue measure, and give F the right $C([0, 1])$ -module action*

$$x \cdot a = xa.$$

Then F is an essential Banach $C([0, 1])$ -module and

$$K_\omega^F = F \quad (\omega \in [0, 1]). \quad (2)$$

Hence every fiber Ξ_ω^F is zero, $\tilde{F} = \{0\}$, and the canonical map ι_F is the zero map. In particular, it is not injective.

Proof. The action is bounded because

$$\|xa\|_1 \leq \|x\|_1 \|a\|_\infty.$$

It is unital, so the Banach module is essential.

Fix $\omega \in [0, 1]$. For $m \geq 1$, define

$$a_m(t) = \min\{1, m|t - \omega|\}, \quad 0 \leq t \leq 1.$$

Another important point in Theorem 4.2 is the automatic $C_0(\Omega)$ -linearity. In fact, it can be shown that for every C^* -algebra A , any bounded local linear map from a Banach right A -module into a Hilbert A -module is automatically A -linear (see Proposition A.1 in the Appendix). Theorem 4.2 tells us that if A is commutative, then one can relax the assumption of the range space to a Banach A -convex module and one can remove the boundedness assumption. Another application of this theorem is that if A is a finite dimensional C^* -algebras, then every local linear map between any two Banach right A -modules is A -linear.

Corollary 4.9. *Let A be a finite dimensional C^* -algebra. Suppose that E and F are unital Banach right A -modules. If $\theta : E \rightarrow F$ is a local $\mathbb{C}$ -linear map in the sense of Proposition A.1 (not assumed to be bounded), then θ is an A -module map.*

Proof: Pick any $x \in E$ and $a \in A_{sa}$. Let $A_a := C^*(a, 1)$. By Remark 2.5(c), both E and F are unital Banach A_a -convex modules. Thus, Theorem 4.2 tell us that θ is a A_a -module map. In particular, $\theta(xa) = \theta(x)a$. $\square$

Remark 4.10. (a) Suppose that A is a unital C^* -algebra and F is a unital Banach right A -convex module in the sense $\|xa + y(1-a)\| \leq \max\{\|x\|, \|y\|\}$ for $x, y \in F$ and $a \in A_+$ with $a \leq 1$. Then, by the argument of Corollary 4.9, all local linear maps from any unital Banach right A -module into F are automatically A -linear.

(b) If one can show that for every compact subset $\Omega \subseteq \mathbb{R}$ and every essential Banach $C(\Omega)$ -module F , the map $\sim : F \rightarrow F$ is injective, then using the argument of Corollary 4.9, one can show that for each C^* -algebra A , all local linear maps between any two Banach right A -modules are A -module maps (without assuming that θ is bounded). However, we do not know if it is true.

Figure 1: The injectivity question in Remark 4.10(b) of arXiv:1005.4561, p. 13.

Then $a_m \in C([0, 1])$, $a_m(\omega) = 0$, and $0 \leq a_m \leq 1$. Thus $a_m \in K_\omega$. Moreover, $a_m(t) \rightarrow 1$ at every $t \neq \omega$. For arbitrary $x \in L^1([0, 1])$,

$$|x(t)a_m(t) - x(t)| \leq |x(t)|.$$

Also, $x(t)a_m(t) \rightarrow x(t)$ for almost every t . Dominated convergence gives $\|xa_m - x\|_1 \rightarrow 0$. Each xa_m belongs to FK_ω , so $x \in \overline{FK_\omega} = K_\omega^F$. Since x was arbitrary, $K_\omega^F = F$.

The argument holds for every ω . Therefore all quotients F/K_ω^F vanish and (1) sends every element to the zero section. As $L^1([0, 1]) \neq \{0\}$, the map is not injective. $\square$

Corollary 2 (atomless L^p family). *Let Ω be a compact Hausdorff space carrying a nonzero finite regular atomless Borel measure μ . For every $1 \leq p < \infty$, the canonical map for the essential module $L^p(\Omega, \mu)$ is identically zero.*

Proof. For fixed ω , regularity and $\mu(\{\omega\}) = 0$ give a decreasing net of neighborhoods U with $\mu(U) \rightarrow 0$. By normality of compact Hausdorff spaces, choose $a_U \in C(\Omega)$ such that $0 \leq a_U \leq 1$, $a_U(\omega) = 0$, and $a_U = 1$ off U . Then for $x \in L^p$,

$$\|x - xa_U\|_p^p \leq \int_U |x|^p d\mu \rightarrow 0.$$

Thus $K_\omega^{L^p} = L^p$ for every ω , exactly as above. $\square$

Scope. Theorem 1 answers the stated injectivity question negatively, and in the strongest possible way: the associated section representation can lose the entire module. Remark 4.10(b) presents injectivity as a sufficient route to proving that arbitrary local maps between Banach right C^* -modules are module maps. The present counterexample blocks that route; it does not by itself decide the broader local-map assertion.

References

- [1] C.-W. Leung, C.-K. Ng, and N.-C. Wong, *Automatic continuity and $C_0(\Omega)$ -linearity of linear maps between $C_0(\Omega)$ -modules*, J. Operator Theory 67 (2012), 3–24; arXiv:1005.4561.